

Literature Status: L_p Compression of $\mathbb{Z} \wr \mathbb{Z}$

Run `fa.banach.001`

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Status. Literature already answered. This packet records a later literature answer to a question in Naor–Peres, *Embeddings of discrete groups and the speed of random walks*, arXiv:0708.0853.

Source question. The source paper asks in Question 7.9, page 21:

Evaluate $\alpha_p^*(\mathbb{Z} \wr \mathbb{Z})$ for $1 < p < 2$.

Here $\alpha_p^*(G)$ denotes the L_p -compression exponent of the finitely generated group G equipped with a word metric.

Supporting answer. Naor–Peres, arXiv:0904.4728, gives the exact value. Its abstract and introduction state

$$\alpha_p^*(\mathbb{Z} \wr \mathbb{Z}) = \max \left\{ \frac{p}{2p-1}, \frac{2}{3} \right\} \quad (1 \leq p < \infty).$$

For the range $1 < p < 2$ asked in the source question,

$$\frac{p}{2p-1} > \frac{2}{3},$$

with equality only at $p = 2$. Hence the answer to Question 7.9 is

$$\alpha_p^*(\mathbb{Z} \wr \mathbb{Z}) = \frac{p}{2p-1} \quad \text{for } 1 < p < 2.$$

Provenance. The supporting paper explicitly says that the computation of $\alpha_p^*(\mathbb{Z} \wr \mathbb{Z})$ answers a question posed in the earlier Naor–Peres paper. This packet is therefore a literature-status record, not a new proof or original partial result.

Scope limitations. Only Question 7.9 is covered here. The other questions in the source paper’s discussion section, including questions about amenable groups, L_1 compression, hyperbolic groups, and edge Markov type, are not addressed by this packet.

References

- [1] A. Naor and Y. Peres, *Embeddings of discrete groups and the speed of random walks*, arXiv:0708.0853.
- [2] A. Naor and Y. Peres, *L_p compression, traveling salesmen, and stable walks*, arXiv:0904.4728; Duke Math. J. 157 (2011), no. 1, 53–108.